
Nutrition Interventions in LMICs

Automated Evidence Synthesis & Prioritization

100

Papers Reviewed

57

Full-Text (PMC)

24

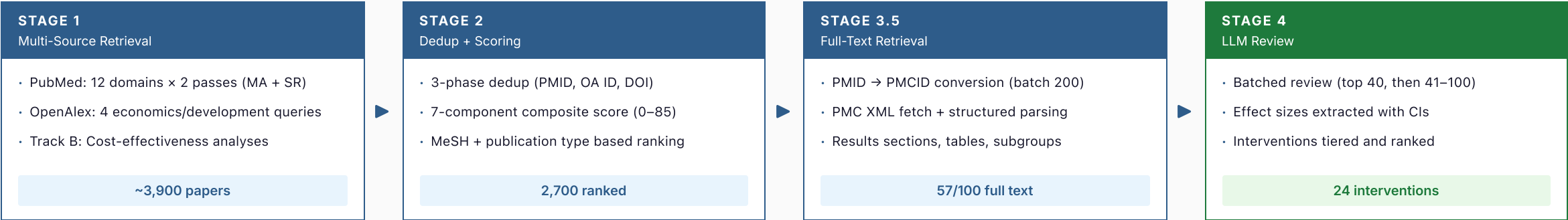
Interventions Ranked

3

Evidence Tiers

PubMed + OpenAlex systematic search · Pipeline v2.0 · May 2026

Automated Evidence Synthesis Pipeline



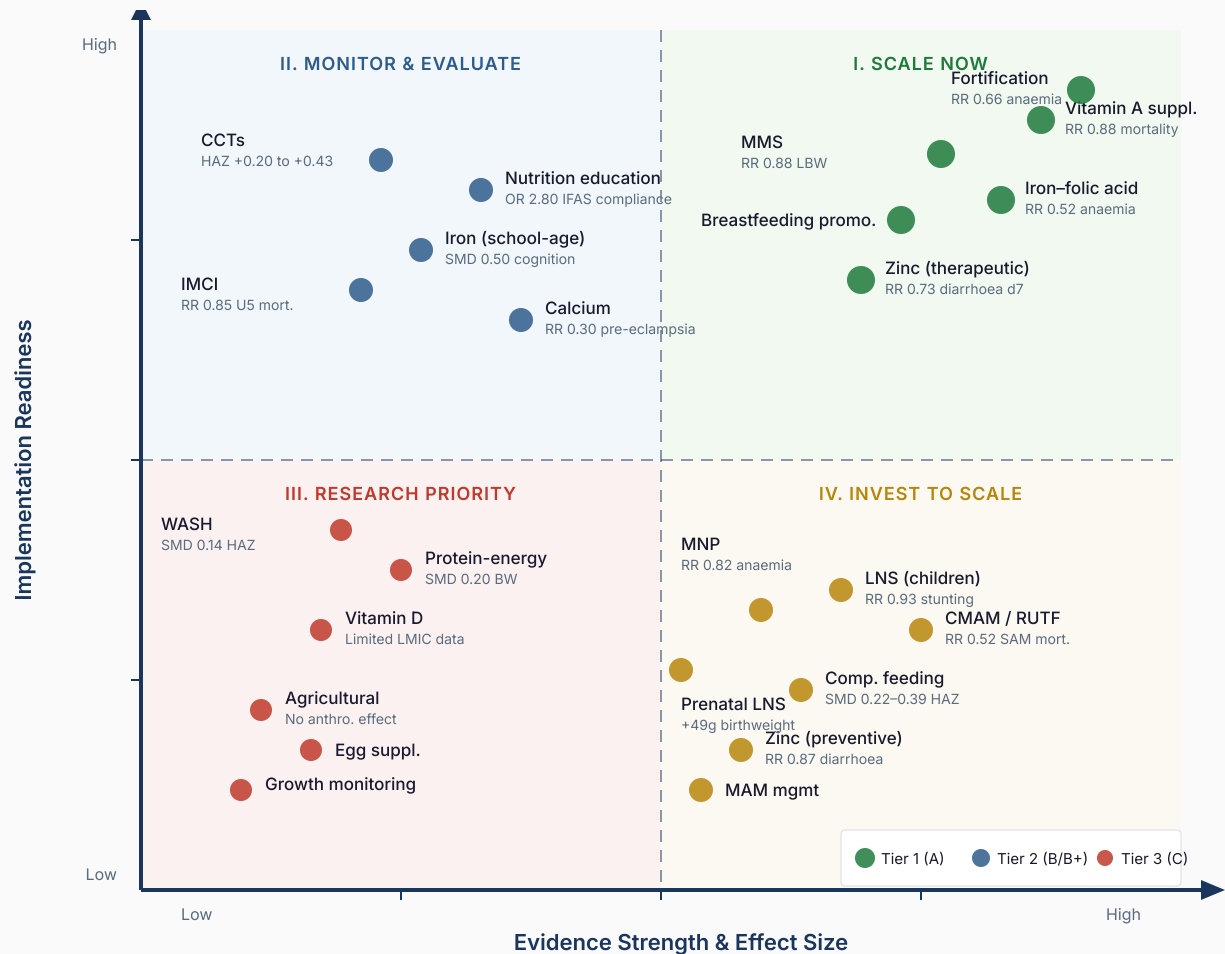
Scoring Components (max 85 points)

Study Design 0–20	Topic Relevance 0–25	Setting 0–10	Recency 0–10	Citation Impact 0–12	Open Access 0–3	Tier Bonus 0–5
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Sources: PubMed E-Utilities (MeSH indexing) · OpenAlex (economics/development) · PMC Open Access (full text) · NCBI ID Converter API

Evidence Prioritization Framework

Figure 1. Twenty-four nutrition interventions positioned by evidence strength and effect size (x-axis) against implementation readiness, a composite of cost-effectiveness and proven scalability (y-axis). Dashed lines delineate four action quadrants. Dot colour indicates evidence tier.



Quadrant Decision Rules

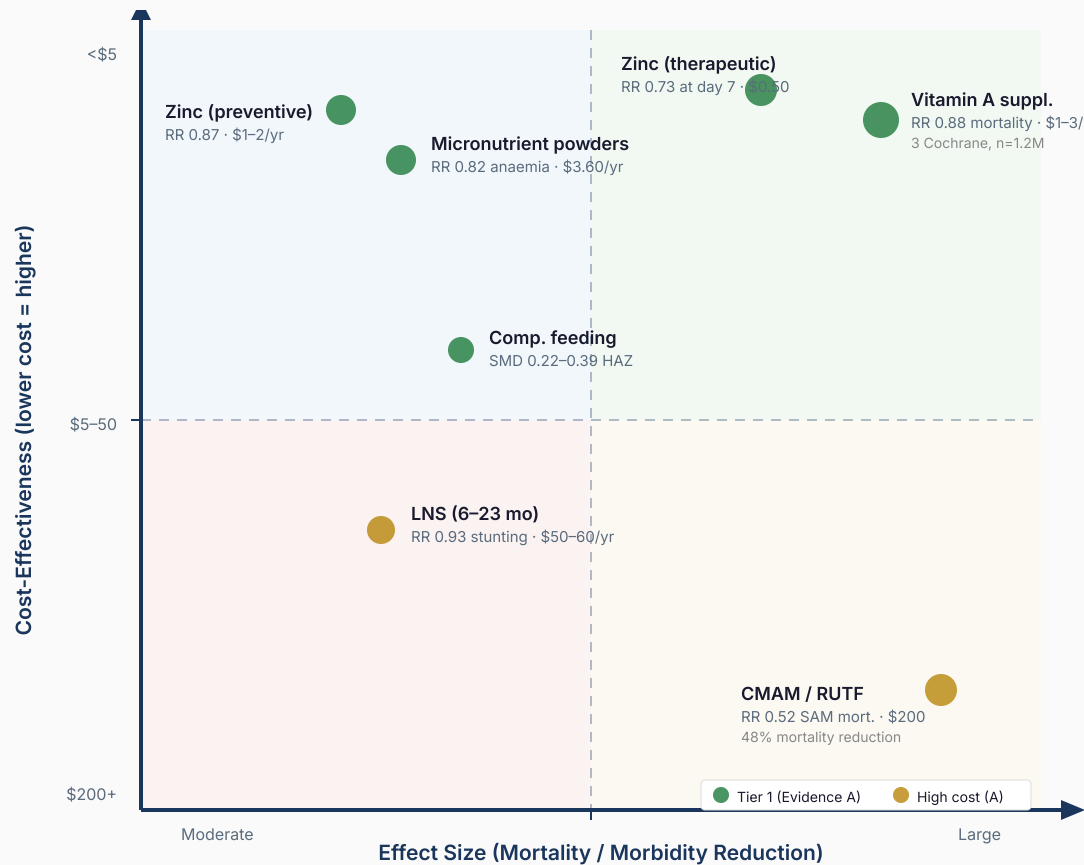
Q	Action	Criteria
I	Scale now	Evidence A + proven national + high cost-effectiveness
II	Monitor & evaluate	Good platforms, moderate evidence (B/B+)
III	Research priority	Emerging evidence (C/C+), insufficient for policy
IV	Invest to scale	Evidence A but higher cost or subnational only

Axis Composition

Axis	Composed of
Evidence (x)	GRADE certainty · # Cochrane reviews · pooled effect magnitude · consistency
Readiness (y)	Cost-effectiveness · delivery platform maturity · # countries with programmes

Children Under 5 — Intervention Prioritization

Figure 2. Child nutrition interventions (6–59 months) positioned by effect size on primary outcome (x-axis) against annual cost per child (y-axis, inverted — lower cost is higher). All interventions shown carry Evidence Rating A. Dot size proportional to evidence base breadth.



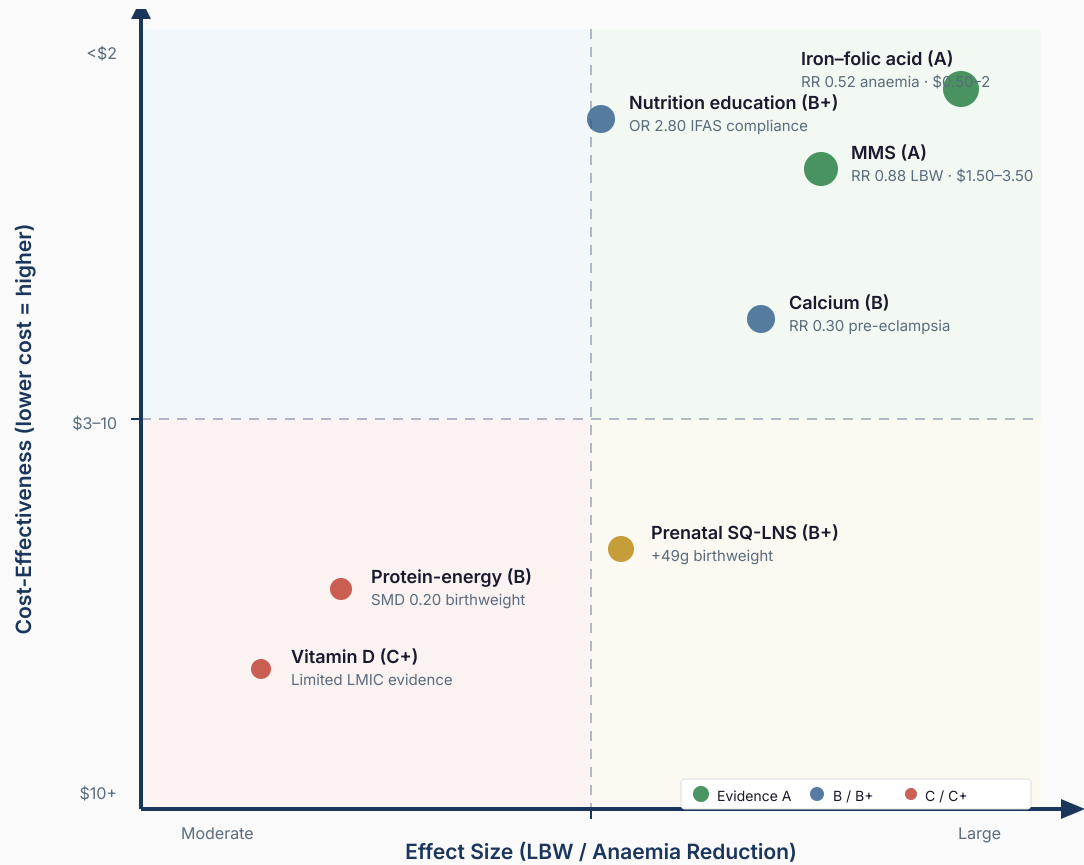
Supporting Data

Intervention	Key Effect	95% CI	Cost
VAS (6–59 mo)	RR 0.88 mortality	0.83–0.93	\$1–3/yr
Zinc (therapeutic)	RR 0.73 at day 7	0.61–0.88	\$0.50
MNP	RR 0.82 anaemia	0.76–0.90	\$3.60/yr
Zinc (preventive)	RR 0.87 diarrhoea	0.85–0.89	\$1–2/yr
Comp. feeding	SMD 0.22 HAZ	0.01–0.43	Variable
LNS (6–23 mo)	RR 0.93 stunting	0.88–0.98	\$50–60/yr
CMAM / RUTF	RR 0.52 mortality	0.43–0.64	\$200

All interventions shown carry Evidence A (multiple Cochrane reviews/meta-analyses). Higher-cost interventions (LNS, CMAM) are justified in high-burden settings where the absolute number of cases averted offsets unit cost.

Pregnant Women — Intervention Prioritization

Figure 3. Maternal nutrition interventions positioned by effect size on primary outcome (x-axis) against cost per pregnancy (y-axis, inverted). Dot colour indicates evidence rating. Evidence ratings shown in parentheses.



Key Policy Insight

MMS costs only **\$1–2 more** than IFA per pregnancy but prevents **12% more low-birthweight births** (RR 0.88 vs IFA alone). WHO conditionally recommended the IFA → MMS transition in 2020.

The Compliance Bottleneck

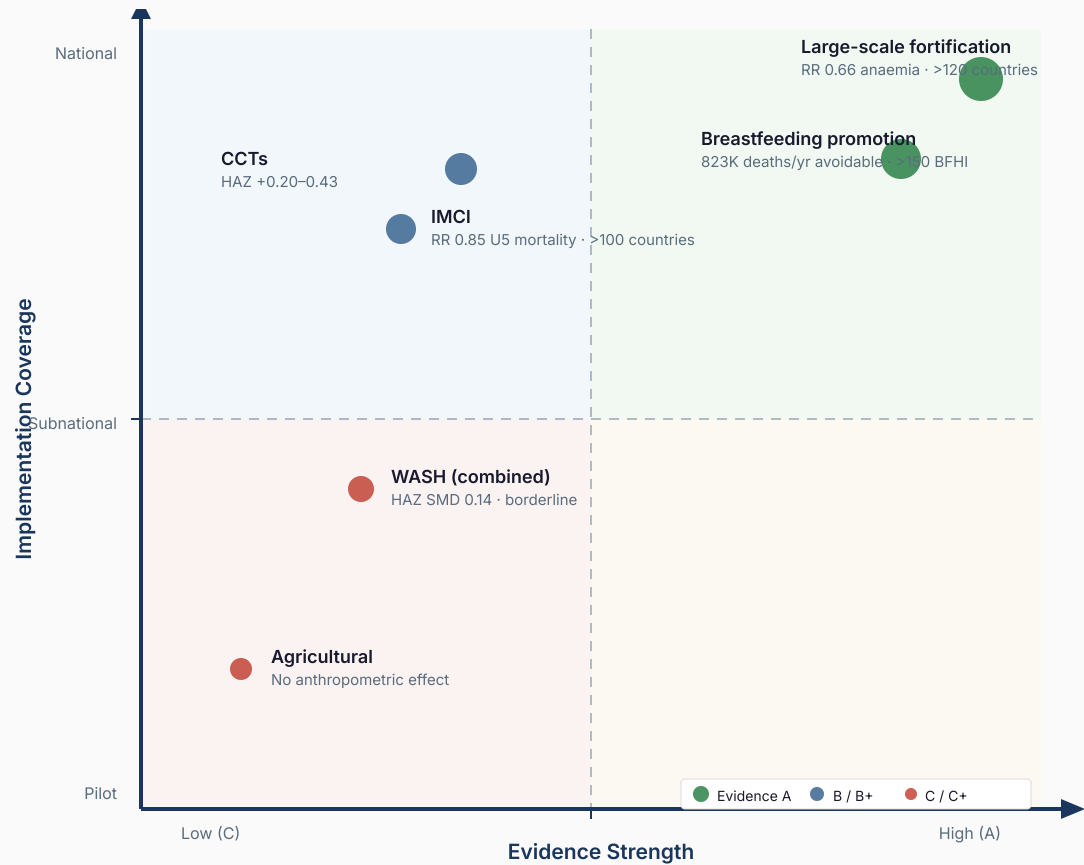
IFA supplement adherence in Sub-Saharan Africa is only **39.2%**. Nutrition education **triples adherence** (OR 2.80) — making it a force-multiplier for any supplementation programme.

Supporting Data

Intervention	Key Effect	95% CI
IFA	RR 0.52 anaemia	0.41–0.66
MMS	RR 0.88 LBW	0.85–0.91
Nutrition ed.	OR 2.80 compliance	2.04–3.83
Calcium	RR 0.30 pre-eclampsia	0.17–0.52
Prenatal LNS	+49g birthweight	IPD, 4 RCTs
Vitamin D	RR 0.40 LBW	0.23–0.71

Population-Level Interventions — Evidence vs. Reach

Figure 4. Population-level and nutrition-sensitive interventions positioned by evidence strength (x-axis) against implementation coverage across LMICs (y-axis). Dot size reflects number of countries with programmes.



Cross-Cutting Finding

76% of studies combining multisectoral policies with nutritional supplementation successfully reduced stunting, vs. variable results for single-sector approaches (PMID 31666032).

The Anaemia Paradox

Iron deficiency explains only **25% of anaemia** in preschool children — far below the commonly assumed 50%. Inflammation, malaria, helminth infections, and other micronutrient deficiencies must be addressed concurrently (PMID 27827838).

Supporting Data

Intervention	Key Effect	Countries
Fortification	RR 0.66 anaemia	>120
Breastfeeding	823K deaths/yr	>150 (BFHI)
CCTs	HAZ +0.20–0.43	Brazil, Mexico, Peru
IMCI	RR 0.85 U5 mort.	>100
WASH	HAZ SMD 0.14	Several (CLTS)
Agricultural	Diet diversity ↑	Growing

Interventions by Delivery Platform & Evidence Tier

Table 5. Classification of all 24 interventions by delivery mechanism and evidence strength. Community-based platforms concentrate the largest number of Evidence A interventions.

Delivery Platform	Evidence A (Strong)	Evidence B / B+ (Moderate)	Evidence C / C+ (Emerging)
Health Facility <i>ANC / Postnatal</i>	Iron–folic acid Multiple micronutrient suppl. Zinc for diarrhoea treatment	Calcium supplementation Nutrition education Iron suppl. (school-age) Prenatal SQ-LNS	Vitamin D supplementation
Community-Based <i>CHW Delivery</i>	Vitamin A supplementation Micronutrient powders SAM management (CMAM) Complementary feeding Breastfeeding promotion Preventive zinc suppl.	WASH interventions Community case mgmt MAM management	Growth monitoring Egg-based supplementation
Food System <i>Fortification / Production</i>	Large-scale fortification (iron, iodine, folic acid)	—	Agricultural interventions Biofortification
Cash / Social Protection	—	Conditional cash transfers IMCI strategy	—

6 of 10 Evidence A interventions are deliverable through community health worker networks — the primary scaling vehicle for child nutrition in LMICs. Health facility–based interventions concentrate in the maternal/pregnancy domain.

Vitamin A Supplementation — What the Pipeline Did Well

Pipeline Strengths

- ✓ Identified **3 separate Cochrane reviews** (2011, 2017, 2022) spanning different eras of evidence
- ✓ Extracted **detailed effect sizes with CIs** from full-text Results sections
- ✓ Distinguished **neonatal VAS** (no benefit; possible harm in Africa) from **6–59 month VAS**
- ✓ Detected **attenuation of effect** in recent trials: RR 0.96 post-2000 vs 0.76 overall
- ✓ Flagged **Asia vs Africa differential**: RR 0.69 vs 0.85 in mortality reduction
- ✓ Identified **adverse effects**: vomiting RR 1.97, bulging fontanelle in <6 mo
- ✓ Noted **declining absolute benefit** as vitamin A deficiency prevalence falls — important for programme reassessment

Effect Sizes Across Cochrane Generations

Outcome	Effect	Period	PMID
All-cause mortality (fixed)	RR 0.88 (0.83–0.93)	2022 Cochrane	35294044
All-cause mortality (random)	RR 0.76 (0.66–0.88)	2017 Cochrane	28282701
Neonatal VAS (overall)	RR 0.97 (0.89–1.06)	IPD, 11 trials	30425075
Neonatal VAS — Asia	RR 0.87 (0.77–0.98)	Subgroup	30425075
Neonatal VAS — Africa	RR 1.06 (0.98–1.15)	Subgroup	30425075
Post-2000 trials only	RR 0.96 (0.89–1.03)	Temporal	35584136
Diarrhoea mortality	RR 0.88 (0.79–0.98)	2022 Cochrane	35294044

The pipeline successfully synthesized evolving evidence across 3 Cochrane generations, detecting both the declining effect trend and critical population-specific signals without human guidance.

Vitamin A Supplementation — Pipeline Limitations

Structural Gaps

- ✗ **CEA blind spot:** Rated "Very High" cost-effectiveness but zero cost-effectiveness papers were retrieved
- ✗ **External knowledge leak:** "\$1–3/child/yr" comes from LLM training data, not from any paper in the database
- ✗ **Misattribution:** "823,000 deaths preventable" cited from PMID 26869575 — but that is the Lancet *Breastfeeding* Series
- ✗ **Statistical model tension:** Fixed-effect RR 0.88 vs random-effects RR 0.76 not flagged as methodological divergence
- ✗ **Overlapping studies:** Cannot detect shared primary trials across the 3 Cochrane reviews
- ✗ **Publication bias:** Cannot assess funnel plot asymmetry
- ✗ **No alternatives:** Biofortification and dietary diversification not evaluated as VAS substitutes

The CEA Blind Spot

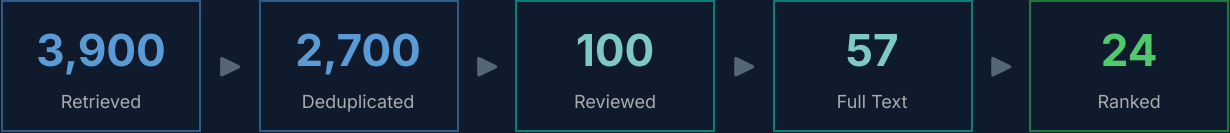
The search included a dedicated cost-effectiveness track (Track B) but retrieved **zero usable papers** for VAS. All cost claims rely on LLM external knowledge. The synthesis acknowledges this but the confidence of the "Very High" rating does not reflect the gap.

The Statistical Model Tension

The 2022 Cochrane used **fixed-effect** (RR 0.88); the 2017 used **random-effects** (RR 0.76). These are different statistical approaches on overlapping primary data. A human reviewer would note: the true effect likely lies between these estimates, and the choice of model matters more than additional trials.

These are structural pipeline gaps, not LLM reasoning failures. Fixes require dedicated cost-effectiveness databases (e.g., Tufts CEA Registry), primary trial registry matching, and explicit statistical model comparison.

Summary & Key Takeaways



Top 5 "Scale Now" Interventions

#	Intervention	Key Effect (95% CI)	Cost	Evidence
1	Vitamin A suppl. (6–59 mo)	RR 0.88 all-cause mortality (0.83–0.93)	\$1–3/child/yr	A — 3 Cochrane
2	Iron–folic acid (pregnancy)	RR 0.52 maternal anaemia (0.41–0.66)	\$0.50–2/pregnancy	A
3	Multiple micronutrient suppl.	RR 0.88 low birthweight (0.85–0.91)	\$1.50–3.50/pregnancy	A — Cochrane
4	Large-scale fortification	RR 0.66 anaemia (0.59–0.74)	\$0.05–0.50/person/yr	A
5	Exclusive breastfeeding promo.	823K under-5 deaths preventable/yr	Very low per beneficiary	A

Key Limitations

- ▶ **No cost-effectiveness papers retrieved** despite dedicated search track — all cost ratings rely on LLM external knowledge
- ▶ **43/100 papers abstract-only** — effect sizes not independently verified from full text for nearly half the evidence base
- ▶ **Geographic concentration** in South Asia and East/West Africa — Latin America, Central Asia, Pacific Islands underrepresented